

TELECOM CUSTOMER CHURN & RETENTION ANALYSIS

Professional End-to-End Data Analytics Project Report

Excel • Power Query • Power BI • MySQL

Project component	Implementation
Source & preparation	Excel workbook with source-stage Master_Data data and final Table1 Database source: Kaggle — Telco Customer Churn IBM Dataset
Data transformation	Power Query — Cleaned_Data query
Analytical model	Power BI fact/dimension structure with dedicated measure tables
Dashboard	Overview + Retention Analysis pages; interactive slicers and navigation
SQL validation	MySQL 8.0+ single-table analysis with KPI, segmentation, risk and value queries
Business focus	Churn exposure, retention, revenue at risk, customer value, services, contracts, payments and lifecycle

Prepared from the supplied Excel workbook, Power BI PBIX, SQL script and dashboard / Power Query screenshots; database source: Kaggle — Telco Customer Churn IBM Dataset.

1. Executive Summary

This project analyzes 7,043 telecom customers to understand churn, retention, recurring-revenue exposure and customer-value patterns. The final analytical table contains 37 fields and is used as the cleaned reporting foundation. Power Query creates business-ready classifications, Power BI turns those fields into an interactive executive dashboard, and MySQL provides an independent validation and analysis layer.

The customer base contains 5,174 retained customers and 1,869 churned customers. Overall churn is 26.5%, leaving a retention rate of 73.5%. Total monthly charges are \$456.1K, of which \$139.1K is associated with churned customers — 30.5% of total monthly charges. Average CLTV is \$4.4K, and the dataset contains 2,983 High Risk customers.

Executive KPI Snapshot

KPI	Value	Interpretation
Total Customers	7,043	Customer population
Retained Customers	5,174 (73.5%)	Customers still retained
Churned Customers	1,869 (26.5%)	Customers who churned
Monthly Revenue / Charges	\$456.1K	Total monthly recurring charges
Revenue Lost	\$139.1K	Monthly charges from churned customers (30.5% of total)
Average CLTV	\$4.4K	Average customer lifetime value
At-Risk Customers	2,983	Customers classified High Risk
Dashboard Date As Of	26 Aug 2026	Displayed in supplied Power BI dashboard

Headline Findings

- New customers have the highest churn rate at 52.9%; the rate declines steadily across the lifecycle to 6.6% for Veteran customers.
- Month-to-month contracts have the highest churn rate at 42.7%, versus 2.8% for two-year contracts.
- Electronic check has the highest payment-method churn rate at 45.3%.
- Fiber optic customers churn at 41.9%, substantially above DSL (19.0%) and customers with no internet service (7.4%).
- High Risk customers account for 2,983 customers and have a 59.2% churn rate.
- Veteran customers have the highest average CLTV at \$5.2K.

2. Report Structure

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3. Project Definition & Business Questions

The analytical objective is to identify where customer churn is concentrated, quantify its recurring-revenue impact, understand lifecycle and service behavior, and prioritize retention opportunities. The project combines descriptive customer analytics with a rule-based Churn_Risk classification and high-value customer analysis.

Business Questions

- How many customers are retained versus churned, and what monthly revenue is exposed?
- Which tenure groups have the greatest churn and the strongest customer value?
- Which contract and payment methods are associated with higher churn?
- How does churn vary across internet and supporting service adoption?
- Which customer demographics show meaningful churn differences?
- Which cities have the largest retained customer populations?
- What reasons are most frequently stated by churned customers?
- How many customers are classified High Risk, and how does their churn compare with other risk groups?
- Which customers or customer groups combine high CLTV or monthly charges with churn exposure?
- Can SQL independently validate the core Power BI KPIs and segmentation outputs?

Analytical Objectives

- Create a repeatable cleaned customer table in Power Query.
- Engineer tenure, charge, CLTV, service, churn-risk and service-category fields.
- Build an interactive Power BI reporting layer with executive and retention views.
- Use SQL for structural QA, KPI validation, segmentation and reusable customer outputs.
- Translate churn patterns into targeted retention actions.

4. Dataset & Workbook Assessment

The supplied workbook contains three worksheets. 'Master_Data' contains 7,045 rows and 33 source-stage columns; 'Cleaned_Data' contains the final cleaned population of 7,043 rows and 37 analytical columns;

Worksheet	Observed size	Role
Master_Data	7,045 × 33	Source-stage dataset; includes Count and Lat Long fields and two trailing non-customer rows
Cleaned_Data	7,043 × 37	Final cleaned analytical table loaded as the reporting dataset

Source-to-Cleaned Population

The source-stage sheet contains 7,043 customer records plus a blank trailing row and a footer-like row containing 7,043 in the Count column. The final Cleaned_Data contains exactly 7,043 unique Customer_ID values, indicating that the non-customer trailing rows were excluded before the final analytical output.

Final Analytical Field Families

Family	Examples
Customer & geography	Customer_ID, Country, State, City, Zip_Code, Latitude, Longitude
Demographics	Gender, Senior_Citizen, Partner, Dependents
Lifecycle	Tenure_Months, Cust_Tenure_Group
Services	Phone_Service, Multiple_Lines, Internet_Service, Online_Security, Online_Backup, Device_Protection, Tech_Support, Streaming_TV, Streaming_Movies
Service segmentation	Service_Count, Service_Category
Commercial	Contract, Paperless_Billing, Payment_Method
Charges & value	Monthly_Charges, Monthly_Charges_Group, Total_Charges, CLTV, CLTV_Group
Churn	Churn_Score, Churn_Risk, Churn_Label, Churn_Value, Churn_Reason

5. End-to-End Architecture

Stage	Tool	Implementation	Output
1. Source	Excel	Master_Data source worksheet	Raw customer attributes
2. Preparation	Power Query	Cleaned_Data query, type changes, derived groups, null handling, duplicate removal	Cleaned_Data / 7,043 × 37
3. Modeling	Power BI	Fact customer table, dimensions for churn, contract, customer, location, service and measure tables	Semantic model
4. Reporting	Power BI	Overview + Retention Analysis pages, KPI cards, charts, map, slicers and navigation	Interactive dashboard
5. Validation	MySQL	Schema, import, QA, KPI, segmentation, risk, value and advanced queries	Independent SQL outputs
6. Decision support	Analysis	Cross-layer reconciliation and business interpretation	Retention recommendations

Design Principle

The solution separates preparation from reporting: Power Query produces standardized analytical fields; Power BI provides the semantic and presentation layer; SQL is used as a separate analytical/validation layer rather than as an unnecessary data-modeling join layer. The supplied SQL explicitly identifies the project as single-table analysis, so no JOIN is required.

6. Excel & Power Query Data Preparation

The supplied Power Query screenshot shows a query named Cleaned_Data with a 37-column, 7,043-row preview. The transformation sequence includes data typing, lifecycle and commercial segmentation, churn-risk construction, service counting/category assignment, text cleanup, column removal/reordering, duplicate removal, and null replacement.

Transformation area	Observed implementation
Column order / structure	Reordered Columns 1; Reordered Columns 2
Data types	Changed Data Type 1; Changed Data Type 2
Lifecycle	Added Tenure_Group
Charges	Added Monthly_Charges_Group
Customer value	Added CLTV_Group
Churn risk	Added Churn_Risk
Service breadth	Added Service_Count
Service segmentation	Added Service_Category
Text quality	Trimmed Text
Column pruning	Removed Columns
Duplicate control	Removed Duplicates
Row filtering	Filtered Rows
Null handling	Replaced Total Charges null (0); Replaced Reason null (N/A)

The visible formula bar on the final replacement step shows Churn_Reason null values being replaced with 'Not Applicable'. This is appropriate for a reason field where non-churned customers do not have a churn reason.

7. Power Query Applied Steps

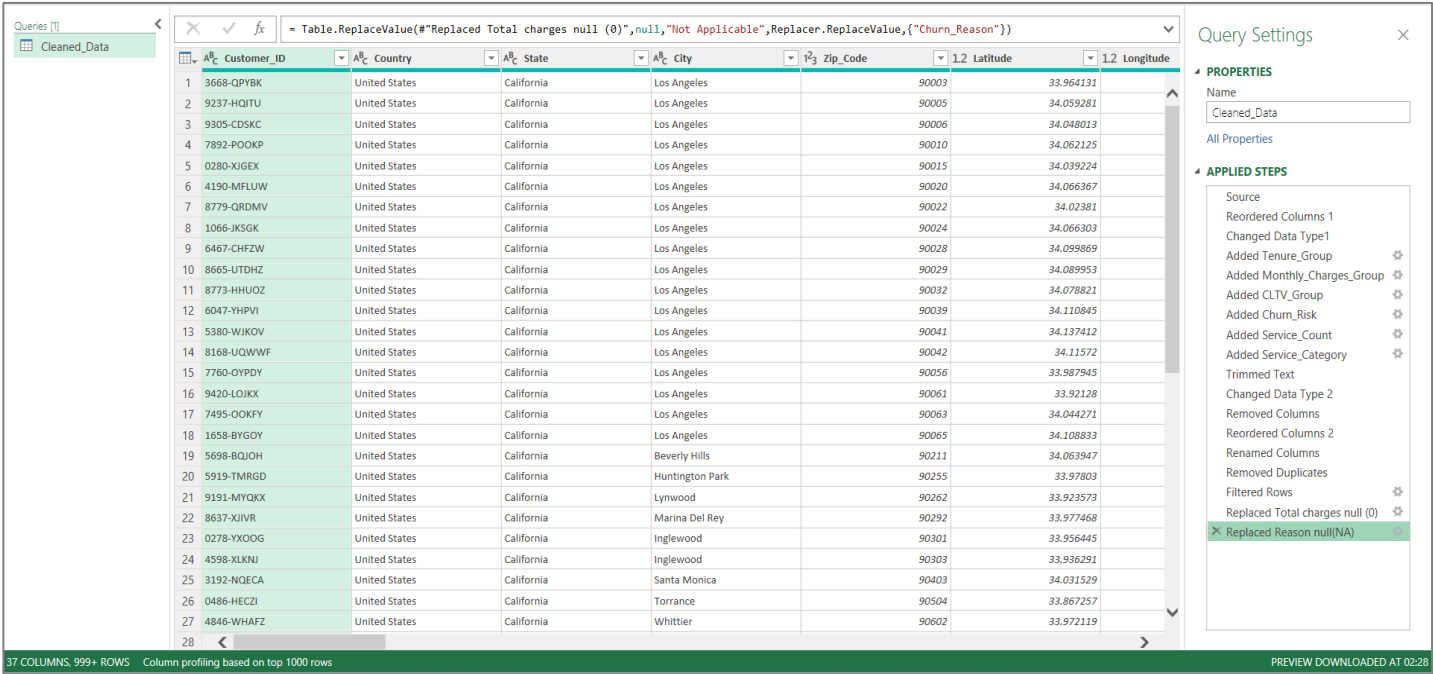


Figure 1. Supplied Power Query evidence — Cleaned_Data query, 37-column/999+ row preview and Applied Steps.

The screenshot provides direct evidence of the repeatable preparation pipeline. The visible final output contains standardized Customer_ID, geographic, tenure and analytical fields, while the Applied Steps pane documents the sequence used to reach the reporting table.

Observed Applied Steps in Sequence

No.	Applied Step
1	Source
2	Reordered Columns 1
3	Changed Data Type 1
4	Added Tenure_Group
5	Added Monthly_Charges_Group
6	Added CLTV_Group
7	Added Churn_Risk
8	Added Service_Count
9	Added Service_Category
10	Trimmed Text
11	Changed Data Type 2
12	Removed Columns
13	Reordered Columns 2
14	Renamed Columns
15	Removed Duplicates
16	Filtered Rows
17	Replaced Total charges null (0)
18	Replaced Reason null (N/A)

8. Data Quality & Reconciliation

Quality check	Result	Interpretation
Final rows	7,043	Matches expected customer population
Final columns	37	37-field analytical table
Unique Customer_ID	7,043 / 7,043	Unique customer key
Duplicate Customer_ID rows	0	No duplicate customer IDs in final table
Missing values in Table1	0	No missing cells observed in final analytical table
Churn Label vs Churn Value	Reconciles	Churn_Value 1 corresponds to Yes; 0 to No
Churn Reason	5174 Not Applicable	Retained customers have no churn reason; churned customers have stated reasons

Important Data-Quality Observation

The raw source worksheet has two trailing non-customer rows, while the final Cleaned_Data has 7,043 unique customers. This is a meaningful preparation control because the SQL import and Power BI reporting use the 7,043-customer analytical population.

9. Power BI Model & Report Architecture

The PBIX metadata shows four report pages/containers, with the two primary analytical pages named Overview and Retention Analysis in the file metadata. The visible dashboard screenshots identify the business-facing views as the Overview and Retention Analysis pages. The model contains a central Fact_Customer_Churn table, dimensions for churn, contract, customer, location and service, plus dedicated _Measures and _Service Measures tables.

Model object	Observed role
Fact_Customer_Churn	Central customer-level analytical fact
Dim_Churn	Churn label / churn risk slicing
Dim_Contract	Contract and payment analysis
Dim_Customer	Gender, Partner, Senior Citizen, Dependents
Dim_Location	City, Latitude, Longitude
Dim_Service	Service category
_Measures	Core customer, churn, revenue, CLTV and percentage measures
_Service Measures	Service-adoption and retained/churned monthly-charge measures
Last Refreshed	Model/report metadata object

Primary Report Pages

Business-facing page	Main purpose
Overview	Executive KPIs, churn by tenure, demographics, service adoption, city distribution and high-level insight
Retention Analysis	Retention by tenure, monthly revenue by tenure, CLTV by tenure, internet service, payment method, contract, service category and retained cities

10. Power BI Measures & Dashboard Design

Core Measures Observed in PBIX Metadata

Measure group	Measures observed
Customer KPIs	Total Customers; Total Active Customers; Total Churn Customers
Revenue	Monthly Charges; Churn Revenue Loss; % Monthly Total; % Churn Revenue Loss
Retention / churn	Active %; Churn %; % of Total
Value / risk	Avg CLTV; At-Risk Cust.
Date / service	Date As Of; %1_OS through %9_PS; churn monthly charges; retained monthly charges
Service churn	%CHURN Service Dis

Dashboard Interaction

- City, Service Category, Contract, Payment Method, Churn Label, Churn Risk and Gender slicers are visible on the dashboard.
- Tenure (Months) is controlled with a range slider from 0 to 72.
- Page navigation separates Overview and Retention Analysis.
- Interactive visual design includes KPI cards, tenure charts, payment/contract comparisons, service adoption, demographic donuts, city tables and a geographic map.

11. Dashboard Evidence — Overview

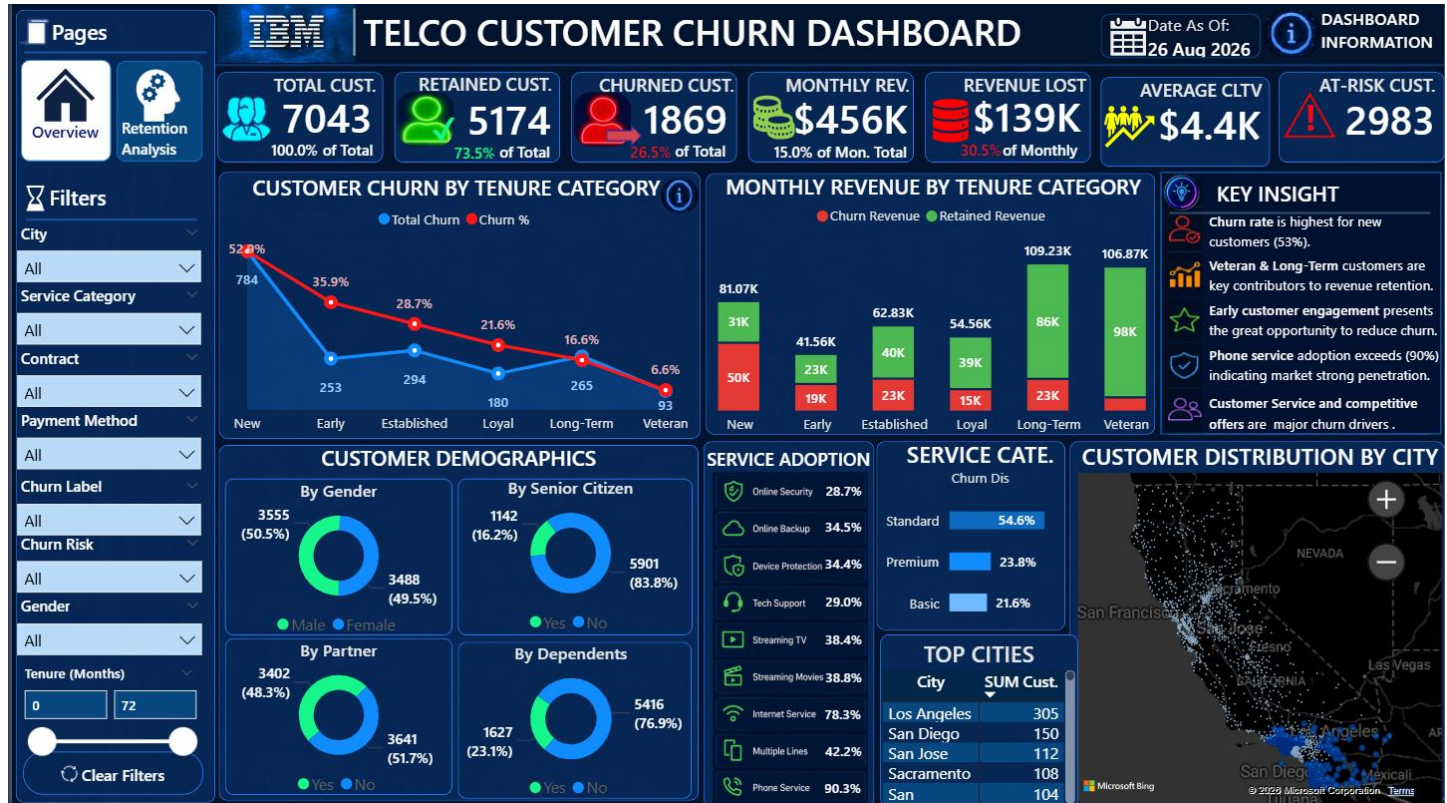


Figure 2. Supplied Power BI Overview dashboard screenshot.

Overview Page Interpretation

- The dashboard establishes the executive baseline with 7,043 total customers, 5,174 retained customers and 1,869 churned customers.
- Monthly revenue/charges are shown at approximately \$456K and churn-associated revenue loss at approximately \$139K.
- Average CLTV is shown at approximately \$4.4K and the dashboard identifies 2,983 at-risk customers.
- The left-side filters allow the entire page to be narrowed by geography, service, contract, payment, churn, risk, gender and tenure.
- The lower dashboard area combines customer demographics, service adoption, service-category mix and geographic distribution.

Dashboard KPI Validation

Dashboard KPI	Underlying Table1 calculation
Total Customers	COUNT(Customer_ID) = 7,043
Retained Customers	Churn_Label = No → 5,174
Churned Customers	Churn_Label = Yes → 1,869
Monthly Revenue	SUM(Monthly_Charges) ≈ \$456,117
Revenue Lost	SUM(Monthly_Charges where Churn_Label = Yes) ≈ \$139,131
Average CLTV	AVG(CLTV) ≈ \$4,400
At-Risk Customers	Churn_Risk = High Risk → 2,983

12. Dashboard Evidence — Retention Analysis

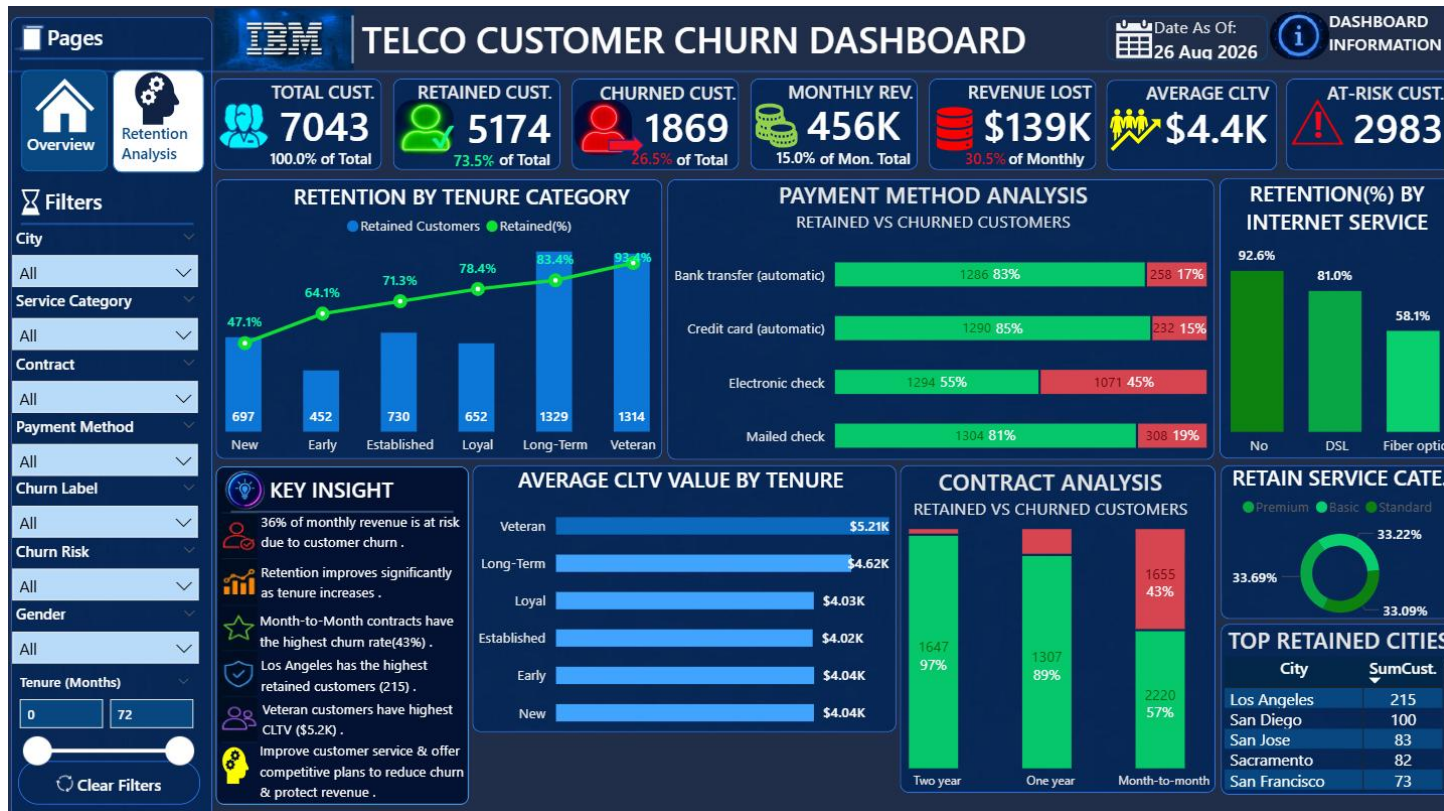


Figure 3. Supplied Power BI Retention Analysis dashboard screenshot.

Retention Analysis View

- Retention rises from 47.1% for New customers to 93.4% for Veteran customers in the displayed tenure analysis.
- Average CLTV increases across the lifecycle, reaching approximately \$5.21K for Veteran customers.
- Internet-service retention is highest for customers with no internet service and lowest for Fiber optic customers in the displayed chart.
- Electronic check shows the weakest payment-method retention, while automatic payment methods perform better.
- Month-to-month contracts show the weakest retention compared with one-year and two-year contracts.
- The dashboard's service-category visual shows the distribution of retained service categories.

The supplied dashboard also includes a Key Insight panel. Its narrative states that 36% of monthly revenue is at risk due to customer churn. The underlying Table1 calculation of churned monthly charges is 30.5% of total monthly charges, while High Risk customers' monthly charges represent 45.1% of total monthly charges. These are different concepts/denominators, so the report preserves the dashboard statement but does not treat the 36% narrative as the same metric as SQL's revenue-lost KPI.

13. Customer Lifecycle & Tenure

Tenure Group	Customers	Churned	Churn Rate	Avg CLTV	Monthly Charges
New	1481	784	52.9%	\$4.0K	\$81.1K
Early	705	253	35.9%	\$4.0K	\$41.6K
Established	1024	294	28.7%	\$4.0K	\$62.8K
Loyal	832	180	21.6%	\$4.0K	\$54.6K
Long-Term	1594	265	16.6%	\$4.6K	\$109.2K
Veteran	1407	93	6.6%	\$5.2K	\$106.9K

	Tenure_Group	Total_Customer	Retained_Customer	Retained_Rate	Churned_Customer	Churned_Rate	Retained_Revenue	Churned_Revenue	Average_CLTV
▶	New Customer	1481	697	47.06	784	52.94	31172	49896	4035
	Early Customer	705	452	64.11	253	35.89	22504	19058	4038
	Established Customer	1024	730	71.29	294	28.71	39748	23082	4015
	Loyal Customer	832	652	78.37	180	21.63	39391	15168	4033
	Long-Term Customer	1594	1329	83.38	265	16.62	86356	22876	4620
	Veteran Customer	1407	1314	93.39	93	6.61	97815	9051	5215

Result 6 ×

Output

Action Output

#	Time	Action	Message
33	12:35:59	SELECT Tenure_Group, COUNT(*) AS Total_Customer, SUM(CASE WHEN Churn_Label='No' THEN...	6 row(s) returned

Figure 4. Independent SQL Validation — churn rate by tenure group.

The lifecycle pattern is pronounced: churn is 52.9% for New customers, falls to 35.9% for Early customers and reaches only 6.6% for Veteran customers. At the same time, average CLTV rises from about \$4.0K for New customers to \$5.2K for Veterans.

14. Contract & Payment Analysis

Contract	Customers	Churned	Churn Rate
Month-to-month	3875	1655	42.7%
One year	1473	166	11.3%
Two year	1695	48	2.8%

	Contract	Total_Customers	Customer_Distribution_Rate	Retained_Customer	Retained_Rate	Churned_Customer	Churned_Rate
▶	Month-to-month	3875	55.02	2220	57.29	1655	42.71
	One year	1473	20.91	1307	88.73	166	11.27
	Two year	1695	24.07	1647	97.17	48	2.83

Result 8 ×

Output

Action Output

#	Time	Action	Message
35	12:37:54	SELECT Contract, COUNT(*) AS Total_Customers, ROUND(COUNT(*)*100/(SELECT COUNT(*) ...	3 row(s) returned

Figure 5. Independent SQL Validation — churn rate by contract.

Payment Method	Customers	Churned	Churn Rate
Electronic check	2365	1071	45.3%
Mailed check	1612	308	19.1%
Bank transfer (automatic)	1544	258	16.7%
Credit card (automatic)	1522	232	15.2%

	Payment_Method	Total_Customers	Total_Cust_Rate	Retained_Customer	Retained_Rate	Churned_Customer	Churned_Rate
▶	Electronic check	2365	33.6	1294	54.71	1071	45.29
	Mailed check	1612	22.9	1304	80.89	308	19.11
	Bank transfer (automatic)	1544	21.9	1286	83.29	258	16.71
	Credit card (automatic)	1522	21.6	1290	84.76	232	15.24

Result 10 ×

Output

Action Output

#	Time	Action	Message
37	12:39:02	SELECT Payment_Method, COUNT(*) AS Total_Customers, ROUND(COUNT(*)*100/7043,1) Tota...	4 row(s) returned

Figure 6. Independent SQL Validation — churn rate by payment method.

Commercial structure is one of the strongest churn differentiators in the dataset. Month-to-month customers have a 42.7% churn rate versus 11.3% for one-year and 2.8% for two-year contracts. Electronic check customers have a 45.3% churn rate, materially above automatic bank transfer (16.7%) and automatic credit card (15.2%).

15. Service Adoption & Churn

Internet Service	Customers	Churned	Churn Rate	Retention Rate
No	1526	113	7.4%	92.6%
DSL	2421	459	19.0%	81.0%
Fiber optic	3096	1297	41.9%	58.1%

Service Adoption Snapshot

Service	Adoption rate
Phone Service	90.3%
Internet Service	78.3%
Online Security	28.7%
Online Backup	34.5%
Device Protection	34.4%
Tech Support	29.0%
Streaming TV	38.4%
Streaming Movies	38.8%
Multiple Lines	42.2%

The dashboard displays Phone Service adoption at 90.3% and Internet Service adoption at 78.3%. Fiber optic is the largest churn exposure among internet-service categories: 1,297 of 3,096 Fiber optic customers churned, a 41.9% rate.

16. Customer Demographics

Gender

Gender	Customers	Churned	Churn Rate
Female	3488	939	26.9%
Male	3555	930	26.2%

Senior Citizen

Senior Citizen	Customers	Churned	Churn Rate
No	5901	1393	23.6%
Yes	1142	476	41.7%

Partner

Partner	Customers	Churned	Churn Rate
No	3641	1200	33.0%
Yes	3402	669	19.7%

Dependents

Dependents	Customers	Churned	Churn Rate
No	5416	1763	32.6%
Yes	1627	106	6.5%

The screenshot shows a data validation tool interface. At the top, there is a table with 11 columns: Category, Response, Total_Customer, Retained_Customer, Retained_Rate, Churned_Customer, Churned_Rate, Retained_Revenue, Churned_Revenue, and Average_CLTV. The table contains 6 rows of data for Gender, Senior_Citizen, and Partner categories. Below the table, there is a section labeled 'Result 13' with a close button. Underneath, there is an 'Output' section with a dropdown menu set to 'Action Output'. At the bottom, there is a table with 4 columns: #, Time, Action, and Message. The first row shows a successful action at 12:41:22, with the message 'SELECT 'Gender' AS Category, Gender AS Response, COUNT(*) AS Total_Customer, SUM(CAS... 8 row(s) returned'.

Category	Response	Total_Customer	Retained_Customer	Retained_Rate	Churned_Customer	Churned_Rate	Retained_Revenue	Churned_Revenue	Average_CLTV
Gender	Female	3488	2549	73.08	939	26.92	157184	70249	4408
Gender	Male	3555	2625	73.84	930	26.16	159802	68882	4392
Senior_Citizen	No	5901	4508	76.39	1393	23.61	264251	100711	4402
Senior_Citizen	Yes	1142	666	58.32	476	41.68	52735	38420	4392
Partner	Yes	3402	2733	80.34	669	19.66	177185	53390	4569
Partner	No	3641	2441	67.04	1200	32.96	139801	85741	4243

#	Time	Action	Message
40	12:41:22	SELECT 'Gender' AS Category, Gender AS Response, COUNT(*) AS Total_Customer, SUM(CAS...	8 row(s) returned

Figure 7. Independent SQL Validation — Customer Demographics.

Gender differences are small: female churn is 26.9% versus 26.2% for male customers. More meaningful differences appear in life circumstances: Senior Citizens churn at 41.7%, customers without partners at 33.0%, and customers without dependents at 32.6%.

17. Geographic Analysis

The Power BI model uses Latitude and Longitude with City to support a geographic customer-distribution map. The SQL layer groups customer outcomes by City and also calculates average service adoption, retained/churned revenue and average CLTV.

Top city by retained customers	Total Customers	Retained	Churned	Churn Rate
Los Angeles	305	215	90	29.5%
San Diego	150	100	50	33.3%
San Jose	112	83	29	25.9%
Sacramento	108	82	26	24.1%
San Francisco	104	73	31	29.8%
Fresno	64	48	16	25.0%
Long Beach	60	45	15	25.0%
Oakland	52	39	13	25.0%
Bakersfield	40	37	3	7.5%
Stockton	44	32	12	27.3%

Los Angeles has the largest retained-customer count in the supplied analysis and is also the leading city by overall customer population. City-level counts should be interpreted with population size in mind; a high churn rate in a small city is not directly comparable with a high churn count in a large city.

18. Churn Reasons & Risk

Top stated churn reason	Churned Customers	Share of churned customers
Attitude of support person	192	10.3%
Competitor offered higher download speeds	189	10.1%
Competitor offered more data	162	8.7%
Don't know	154	8.2%
Competitor made better offer	140	7.5%
Attitude of service provider	135	7.2%
Competitor had better devices	130	7.0%
Network reliability	103	5.5%
Product dissatisfaction	102	5.5%
Price too high	98	5.2%

Risk Distribution

Risk	Customers	Churned	Churn Rate	Customer Share
High Risk	2983	1767	59.2%	42.4%
Medium Risk	2868	102	3.6%	40.7%
Low Risk	1192	0	0.0%	16.9%

The rule-based risk field is highly discriminative in the supplied data: High Risk customers churn at 59.2%, Medium Risk at 3.6%, and Low Risk at 0.0%. The High Risk group contains 2,983 customers and 1,767 churned customers.

Top Churn Themes

- Service/support experience: attitude of support person and attitude of service provider are among the leading stated reasons.
- Competitive pressure: higher download speeds, more data, better offers and better devices are prominent reasons.
- Product/network experience: network reliability and product dissatisfaction appear in the top reasons.
- Price/value pressure: price too high and extra data charges are recurring themes.

19. Customer Value & Revenue Exposure

CLTV Group	Customers	Churned	Churn Rate	Avg CLTV	Monthly Charges
Low CLTV	1179	423	35.9%	\$2.5K	\$71.3K
Medium CLTV	2285	642	28.1%	\$3.9K	\$144.5K
High CLTV	3579	804	22.5%	\$5.4K	\$240.4K

Monthly-Charge Exposure

Monthly Charges Group	Customers	Churned	Churn Rate	Monthly Charges
Low	1735	189	10.9%	\$38.2K
Medium	1725	413	23.9%	\$94.6K
High	3583	1267	35.4%	\$323.3K

High CLTV customers have the lowest churn rate among the three CLTV groups (22.5%), but they still represent the largest customer-value pool. This makes value-based retention important: a smaller number of churns among high-value customers can carry meaningful commercial impact.

Revenue Exposure

Metric	Value
Total monthly charges	\$456.1K
Churned-customer monthly charges	\$139.1K
Churned charge share	30.5%
High Risk customers' monthly charges	\$205.8K
High Risk monthly-charge share	45.1%

20. SQL Analysis & Validation

The supplied MySQL script is structured as a complete analytical workflow for the cleaned customer table. It explicitly uses a single-table design and therefore does not introduce unnecessary JOIN operations.

Section	SQL scope	Purpose
01	Database & Table Setup	Create database and telco_customer_churn table.
02	Data Import & MySQL Configuration	Enable local_infile and bulk-load cleaned CSV.
03	Data Structure & Quality Validation	Schema inspection, sample rows, row count, duplicate Customer_ID and critical-field checks.
04	Overall Customer & Churn KPI Analysis	Customer counts, retention/churn, monthly revenue, revenue lost, average CLTV and high-risk customers.
05	Tenure & Customer Lifecycle Analysis	Churn, retention, revenue and CLTV by tenure group.
06	Customer Demographic Analysis	Gender, Senior Citizen, Partner and Dependents.
07	Geographic Customer Analysis	City-level customer, churn, service adoption, revenue and CLTV.
08	Service Adoption & Product Analysis	Service category and service-level churn analysis.
09	Contract & Payment Analysis	Contract and payment-method churn.
10	Churn Reason & Churn Risk Analysis	Reason distribution, risk distribution and high-risk above-average-charge customers.
11	Customer Value Analysis	Above-average CLTV and churn status of high-value customers.
12	Customer Charges & Value Segmentation	Monthly charge and CLTV group distributions.
13	Advanced Customer Analysis	Top customers by monthly charges within contract using RANK().
14	Reusable Business View	Create and inspect vw_churned_customers.

SQL Techniques Demonstrated

- DDL: CREATE DATABASE, USE and CREATE TABLE.
- Bulk loading with LOAD DATA INFILE.
- Data-quality controls using COUNT, NULL checks and duplicate detection.
- Conditional aggregation with CASE WHEN for churn/retention KPIs.
- Window functions with SUM(COUNT(*)) OVER() for distribution percentages.
- UNION ALL to standardize multiple service-dimension outputs.
- CTEs for high-value and ranked customer analysis.
- RANK() OVER(PARTITION BY Contract ORDER BY Monthly_Charges DESC) for within-contract ranking.
- Subqueries for above-average CLTV and monthly-charge thresholds.
- CREATE OR REPLACE VIEW for a reusable churned-customer business view.
- Indexes on common analytical columns are included in the performance-optimization portion of the script.

21. Cross-Layer Reconciliation

Metric	Table1 calculation	Power BI evidence	SQL definition
Total customers	7,043	7,043	COUNT(*)
Retained customers	5,174	5,174	CASE Churn_Label='No'
Churned customers	1,869	1,869	CASE Churn_Label='Yes'
Retention rate	73.5%	73.5%	Retained / total
Churn rate	26.5%	26.5%	Churned / total
Monthly charges	\$456.1K	\$456K	SUM(Monthly_Charges)
Revenue lost	\$139.1K	\$139K	SUM(Monthly_Charges where Churn_Label='Yes')
Average CLTV	\$4.4K	\$4.4K	AVG(CLTV)
At-risk customers	2,983	2,983	Churn_Risk='High Risk'

Validation Result

The principal dashboard KPIs reconcile directly to the cleaned Table1 and to the SQL KPI query. Minor differences are only presentation rounding (for example, \$456K versus the calculated \$456,116.60). The dashboard's separate narrative statement about 36% monthly revenue at risk is documented as a dashboard insight rather than substituted for the SQL-defined 30.5% churned-revenue share.

22. Key Business Insights

1. Customer churn is 26.5%, with 1,869 customers churned from a 7,043-customer base.
2. New customers have a 52.9% churn rate, making early lifecycle engagement the strongest retention opportunity.
3. Veteran customers churn at only 6.6% and have the highest average CLTV at about \$5.21K.
4. Month-to-month contracts are the highest-risk commercial structure, with a 42.7% churn rate.
5. Electronic check is the highest-churn payment method at 45.3%.
6. Fiber optic customers have a 41.9% churn rate, the highest among internet-service categories.
7. Senior Citizens churn at 41.7%, compared with 23.6% among non-senior customers.
8. Customers without dependents churn at 32.6%, versus only 6.5% for customers with dependents.
9. High Risk customers have a 59.2% churn rate and represent 2,983 customers.
10. The top stated churn reasons include support-provider attitude, competitive download-speed offers, more-data offers, competitor offers and better devices.
11. Churned customers account for approximately \$139.1K in monthly charges, equal to 30.5% of the total monthly charge base.
12. High CLTV customers churn less frequently than Low CLTV customers, but their larger value pool makes targeted protection commercially relevant.

23. Recommendations

Recommendation	Evidence / rationale	Priority
Create a 0–6 month retention journey	New customers have 52.9% churn; early engagement is the clearest lifecycle opportunity.	High
Prioritize month-to-month customers	42.7% churn versus 2.8% for two-year contracts.	High
Promote contract migration	Use targeted incentives to move suitable month-to-month customers to one-year/two-year plans.	High
Investigate electronic-check friction	Electronic check churn is 45.3%; test billing experience, reminders and automatic-payment migration.	High
Improve Fiber optic experience	Fiber optic churn is 41.9%; investigate price/value, speed, reliability and competitive offers.	High
Strengthen support and service recovery	Support-provider attitude is the top stated churn reason.	High
Target competitive threats	Speed, data, devices and better offers appear repeatedly in stated churn reasons.	High
Protect high-risk/high-value customers	Combine Churn_Risk, CLTV and Monthly_Charges for prioritized outreach.	High
Use service bundling selectively	Service_Count and Service_Category enable adoption-based retention offers.	Medium
Monitor senior and household segments	Senior Citizens and customers without dependents show higher churn.	Medium
Use city-level targeting carefully	Prioritize high-volume cities while separating churn counts from churn rates.	Medium

24. Limitations & Analytical Controls

Limitations

- The dataset represents customer-level observations rather than a full longitudinal event history, so it is primarily descriptive.
- Churn associations should not be interpreted as causal effects without experimental or longitudinal validation.
- Churn_Risk is a rule-based classification created in the analytical dataset; it is not a statistically validated predictive model.
- Churn_Reason is populated for churned customers and 'Not Applicable' for retained customers, so reason analysis applies only to churned customers.
- The supplied PBIX metadata exposes model objects, measure names and visual-query structure, but not every DAX formula as readable plain text.
- The dashboard displays a 'Date As Of' of 26 Aug 2026; this is a report display date, not evidence that customer records are dated on that day.

Controls

- Unique Customer_ID validation.
- Final-table missing-value validation.
- Source trailing-row cleanup.
- Null replacement for Total_Charges and Churn_Reason.
- Duplicate-removal step in Power Query.
- SQL KPI reproduction.
- Cross-layer reconciliation between Excel, Power BI and SQL.

25. Conclusion

The Telecom Customer Churn & Retention Analysis project delivers a complete analytics pipeline over 7,043 customers. The final 37-field analytical table is prepared through Power Query, modeled and presented in Power BI, and independently analyzed and validated through MySQL.

The strongest retention message is lifecycle and commercial: new and month-to-month customers are substantially more likely to churn, while veteran and longer-contract customers show stronger retention and higher CLTV. Service and competitive experience also matter, with Fiber optic and electronic-check customers showing elevated churn and support/competitive issues appearing prominently in stated churn reasons.

The resulting solution is suitable as a portfolio project because it demonstrates data preparation, feature engineering, dimensional modeling, DAX-oriented KPI design, dashboard development, SQL analysis, data-quality validation and business recommendations within one coherent customer-retention workflow.

26. Technical Skills Demonstrated

Skill	Evidence in project
Excel	Source and cleaned customer worksheets; final analytical table
Power Query	18 observed transformation steps including types, grouping, filtering, null handling and duplicate removal
Data Cleaning	Trailing-row exclusion, trimming, type conversion, null replacement and column management
Feature Engineering	Tenure, charges, CLTV, churn-risk, service count and service category
Power BI Modeling	Fact table, churn/contract/customer/location/service dimensions and dedicated measure tables
DAX / Measures	Customer, churn, revenue, percentage, CLTV, risk and service measures
Dashboard Design	KPI cards, tenure charts, stacked comparisons, donuts, service visuals, map, city tables and slicers
SQL	DDL, bulk loading, QA, aggregation, CASE, window functions, CTEs, subqueries, ranking and views
Customer Analytics	Churn, retention, lifecycle, contract, payment, service and customer-value analysis
Business Intelligence	Executive insights, risk prioritization and retention recommendations
Validation	Excel/Cleaned_Data ↔ Power BI ↔ SQL KPI reconciliation

Appendix A — Data Dictionary

Field	Definition	Role
Customer_ID	Unique customer identifier	Key
Country	Customer country	Geography
State	Customer state	Geography
City	Customer city	Geography
Zip_Code	Customer ZIP code	Geography
Latitude	Customer latitude	Geography
Longitude	Customer longitude	Geography
Gender	Customer gender	Demographic
Senior_Citizen	Senior citizen indicator	Demographic
Partner	Partner indicator	Demographic
Dependents	Dependents indicator	Demographic
Tenure_Months	Customer tenure in months	Lifecycle
Cust_Tenure_Group	Business lifecycle category	Derived
Phone_Service	Phone service status	Service
Multiple_Lines	Multiple-line service status	Service
Internet_Service	Internet service type	Service
Online_Security	Online security status	Service
Online_Backup	Online backup status	Service
Device_Protection	Device protection status	Service
Tech_Support	Technical support status	Service
Streaming_TV	Streaming TV status	Service
Streaming_Movies	Streaming movies status	Service
Service_Count	Count of adopted services	Derived
Service_Category	Basic / Standard / Premium service grouping	Derived
Contract	Contract type	Commercial
Paperless_Billing	Paperless billing status	Commercial
Payment_Method	Payment method	Commercial
Monthly_Charges	Monthly recurring charges	Financial
Monthly_Charges_Group	Monthly-charge band	Derived
Total_Charges	Total customer charges	Financial
CLTV	Customer lifetime value	Value
CLTV_Group	CLTV band	Derived
Churn_Score	Source churn score	Churn
Churn_Risk	Rule-based risk category	Derived
Churn_Label	Yes / No churn outcome	Churn
Churn_Value	Numeric churn flag	Churn
Churn_Reason	Stated churn reason; Not Applicable for retained customers	Churn

Appendix B — SQL Query Inventory

ID	Query title	Analytical purpose
01	Database & Table Setup	Create database and telco_customer_churn table.
02	Data Import & MySQL Configuration	Enable local_infile and bulk-load cleaned CSV.
03	Data Structure & Quality Validation	Schema inspection, sample rows, row count, duplicate Customer_ID and critical-field checks.
04	Overall Customer & Churn KPI Analysis	Customer counts, retention/churn, monthly revenue, revenue lost, average CLTV and high-risk customers.
05	Tenure & Customer Lifecycle Analysis	Churn, retention, revenue and CLTV by tenure group.
06	Customer Demographic Analysis	Gender, Senior Citizen, Partner and Dependents.
07	Geographic Customer Analysis	City-level customer, churn, service adoption, revenue and CLTV.
08	Service Adoption & Product Analysis	Service category and service-level churn analysis.
09	Contract & Payment Analysis	Contract and payment-method churn.
10	Churn Reason & Churn Risk Analysis	Reason distribution, risk distribution and high-risk above-average-charge customers.
11	Customer Value Analysis	Above-average CLTV and churn status of high-value customers.
12	Customer Charges & Value Segmentation	Monthly charge and CLTV group distributions.
13	Advanced Customer Analysis	Top customers by monthly charges within contract using RANK().
14	Reusable Business View	Create and inspect vw_churned_customers.

SQL Script Quality Notes

- The script is logically organized from environment setup → import → validation → KPI → segmentation → advanced analysis → reusable view.
- The analysis correctly avoids JOINS because the primary analytical table is single-table.
- Several percentage queries use the fixed denominator 7,043. This is valid for the current dataset, but COUNT(*) or a subquery would be more refresh-safe.
- The SQL file contains a duplicate-style numbering presentation inside section 03 (for example, '03 04' and '03 05'). For a portfolio version, these can be normalized to 03.04 and 03.05 under the same validation section.
- The script uses both '7043' and dynamic COUNT(*) denominators. Standardizing these to dynamic denominators would improve maintainability without changing current results.
- The RANK() query and reusable churned-customer view add advanced SQL depth beyond simple aggregation.

Appendix C — Evidence Register

Artifact	Evidence used in report
Telecom_Customer_Churn_Cleaned_Data.xlsx	Workbook structure, Cleaned_Data , source-stage data, final values, field definitions and data-quality calculations
Telecom_Customer_Churn_Retention_Analysis.pbix	Report pages, visual metadata, model objects, measures and field references
Telecom_Customer_Churn_SQL_Validation.sql	Database/table setup, import, QA, KPI, lifecycle, demographics, geography, service, contract, payment, risk, value, ranking and view analysis
.1_Power_Query_Applied_Steps.png	Power Query Cleaned_Data screenshot and Applied Steps evidence
02_Dashboard_Retention_Analysis.png	Power BI Retention Analysis dashboard screenshot
01_Dashboard_Overview.png	Power BI Overview dashboard screenshot

Final Deliverable Note

This report is based on the supplied project artifacts and preserves the project's own terminology, dashboard framing and SQL organization. Where an analytical discrepancy exists between a dashboard narrative and an independently reproducible calculation, it is explicitly disclosed rather than silently corrected.